

Module 5

Digital Communication Techniques

Syllabus

- *Basic Modulation techniques (AM, FM, PM)*
 - *Multiplexing: FDM, TDM*
 - *Error detection and correction techniques*
-

5.1 Modulation

In telecommunication, the information or 'message' signal (such as voice or music) usually has a low frequency, called the baseband signal. However, this low-frequency signal is unsuitable for direct long-distance transmission using a practical antenna. To overcome this, we use a process called Modulation.

Modulation is defined as the process of varying a parameter (amplitude, frequency, or phase) of a high-frequency carrier signal in accordance with the instantaneous value of the low-frequency information signal.

In simple words, modulation is the technique of 'superimposing information on a high-frequency carrier wave' so that it can be transmitted over long distances.

5.1.1 Key Signal Definitions

- **Modulating Signal (Message Signal):** This is the baseband signal that contains the actual information to be transmitted (e.g., audio from a microphone). It has low frequency and low energy.
- **Carrier Signal:** This is a high-frequency sinusoidal wave produced by an oscillator. It acts as a "carrier" that transports the message through the medium. It contains no information of its own.
- **Modulated Signal:** The resulting wave after the modulation process, which contains the characteristics of both the message and the carrier.

5.1.2 Necessity of Modulation

Modulation is essential in modern communication systems for several technical reasons:

- **Reduction in Antenna Height:** For efficient transmission, an antenna's height must be roughly $\lambda/4$ (one-quarter of the signal wavelength). Low-frequency signals, such as audio signals (20 Hz – 20 kHz), have huge wavelengths ($\lambda = c/f$) and would require antennas kilometers high. Modulation increases the frequency, significantly reducing the antenna size to a practical few meters. For example, a 20 kHz audio signal has a wavelength of 15 km, requiring an antenna of approximately 3.75 km, which is impossible to construct. By using modulation to shift the signal to a higher frequency (say 1 MHz), the required antenna length reduces to a practical 75 meters.
- **Multiplexing:** Multiple message signals can be transmitted simultaneously over the same communication channel by modulating them onto different carrier frequencies. This process, known as frequency division multiplexing (FDM), enables efficient utilization of the available bandwidth. For instance, multiple radio stations can broadcast simultaneously without interfering with each other because each station uses a different carrier frequency.
- **Long-Distance Transmission:** High-frequency modulated signals can travel longer distances through the atmosphere and can be reflected by the ionosphere, enabling long-distance communication. Low-frequency baseband signals cannot achieve such propagation characteristics.
- **Noise Immunity:** Low-frequency signals are more susceptible to noise and interference from various sources. High-frequency carrier signals can propagate through the medium with less noise pickup. Additionally, certain modulation techniques like FM provide better noise immunity compared to direct transmission of baseband signals.

5.1.3 Classification of Analog Modulation

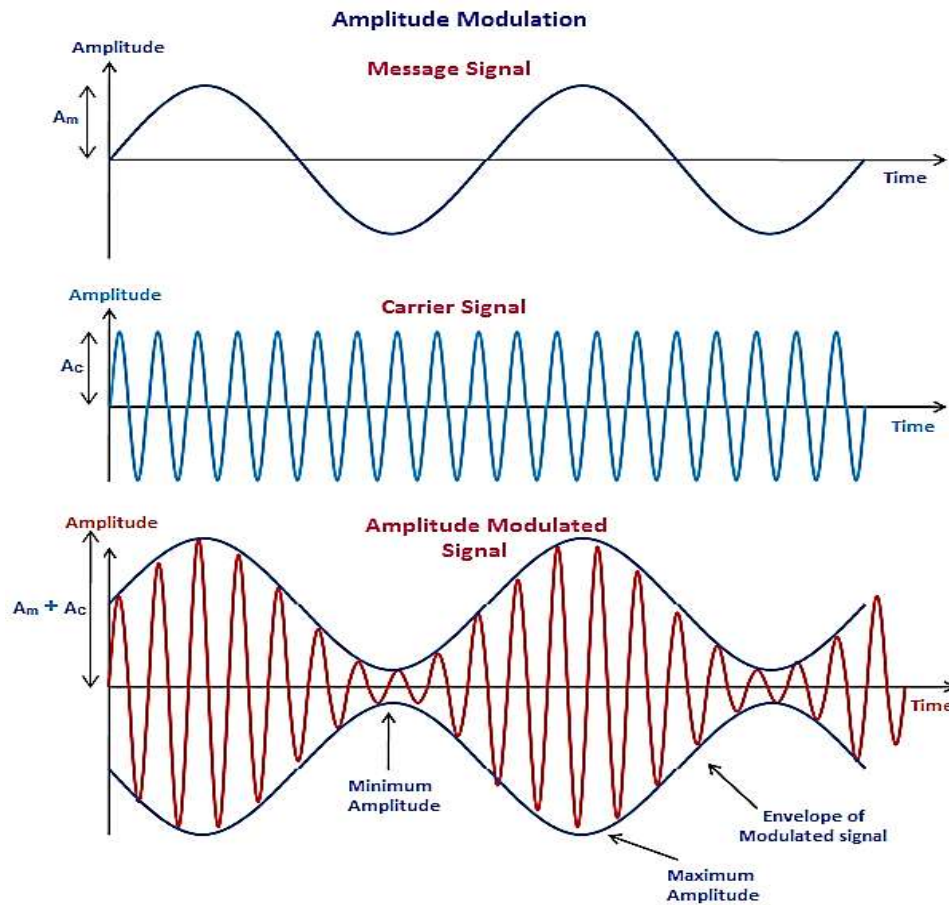
Based on which parameter of the sinusoidal carrier is varied, modulation is classified into:

- Amplitude Modulation (AM) – Carrier amplitude is varied
- Frequency Modulation (FM) – Carrier frequency is varied
- Phase Modulation (PM) – Carrier phase is varied

5.2 Amplitude Modulation (AM)

Amplitude Modulation (AM) is a modulation technique in which the amplitude of the high-frequency carrier signal is varied in accordance with the instantaneous

value of the low-frequency modulating signal, while the frequency and phase of the carrier remain constant. In AM, the information is conveyed through variations in the amplitude of the carrier wave.



5.2.1 Mathematical Representation of AM

Let:

- Carrier signal: $c(t) = A_c \cos(2\pi f_c t)$
- Message signal: $m(t) = A_m \cos(2\pi f_m t)$

The AM signal is given by:

$$s(t) = [A_c + m(t)] \cos(2\pi f_c t)$$

$$s(t) = [A_c + (A_m \cos(2\pi f_m t))] \cos(2\pi f_c t)$$

$$s(t) = A_c [1 + (A_m/A_c) \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

$$s(t) = A_c [1 + \mu \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

where

$$\mu = \frac{A_m}{A_c}$$

is the modulation index.

5.2.2 Modulation Index (Depth of Modulation)

The modulation index (also called depth of modulation or modulation factor) determines the extent to which the carrier amplitude is varied by the modulating signal. It is defined as the ratio of the peak amplitude of the modulating signal (A_m) to the peak amplitude of the carrier signal (A_c).

$$\text{Modulation index } (\mu) = \frac{A_m}{A_c}$$

The modulation index is often expressed as a percentage, called the percentage modulation:

$$\text{Percentage Modulation} = \frac{A_m}{A_c} \times 100\%$$

5.2.3 Significance of Modulation Index

The modulation index is highly significant in AM systems because it directly affects the quality and efficiency of transmission. Based on the value of the modulation index, amplitude modulation can be classified into three categories:

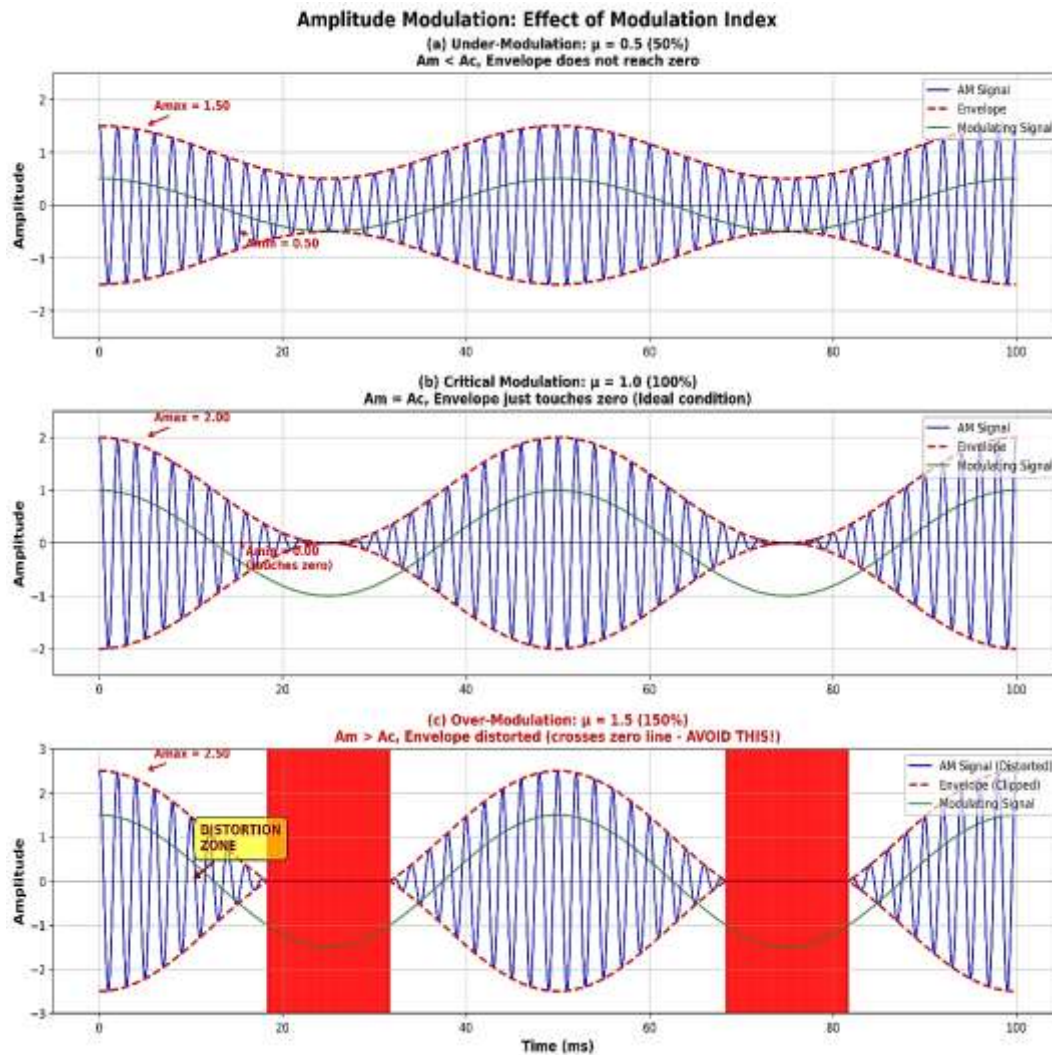
Under-Modulation ($\mu < 1$): When the modulation index is less than 1, the condition is called under-modulation. In this case, the modulating signal amplitude is less than the carrier amplitude ($A_m < A_c$). The AM envelope does not reach zero, and the minimum amplitude $A_{\min}$ is positive. Under-modulation results in inefficient use of transmitter power because a significant portion of power remains in the carrier, which carries no information. However, the signal is free from distortion.

Critical Modulation ($\mu = 1$ or 100% Modulation): When the modulation index equals 1, it is called critical modulation or 100% modulation. In this case, the amplitude of the modulating signal equals the carrier amplitude ($A_m = A_c$). The AM envelope just touches zero at the minimum point ($A_{\min} = 0$), and reaches twice the carrier amplitude at the maximum ($A_{\max} = 2A_c$). This represents the ideal condition for AM transmission, as it provides maximum power efficiency without causing distortion. In practice, systems are designed to operate just below 100% modulation to avoid any risk of over-modulation.

Over-Modulation ($\mu > 1$): When the modulation index exceeds 1, the condition is called over-modulation. Here, the modulating signal amplitude exceeds the carrier amplitude ($A_m > A_c$). This causes the envelope of the AM wave to become distorted, as the minimum amplitude tries to go negative, which is physically impossible. Over-modulation results in severe distortion of the signal and generates additional frequency components (harmonic distortion) that can cause

interference with adjacent channels. Over-modulation must be strictly avoided in AM transmission systems.

Note that, the range of modulation index for distortion-free transmission is $0 \leq \mu \leq 1$. When $\mu = 0$, only the carrier signal is transmitted.



5.2.4 AM Envelope

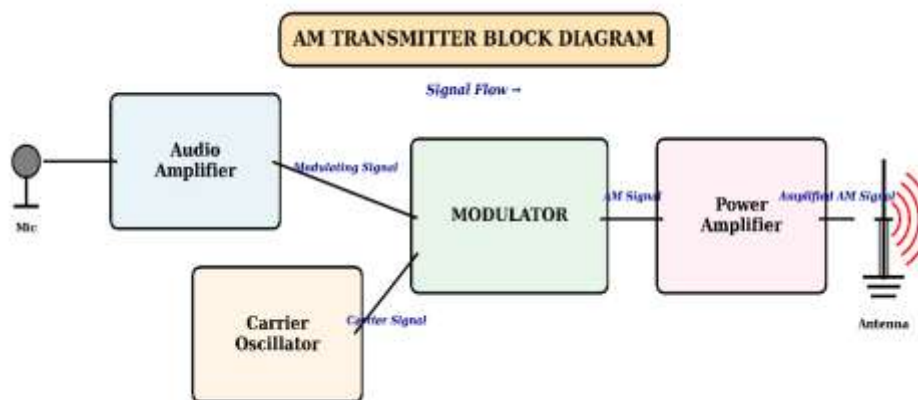
- The imaginary line connecting the peaks of the AM wave is called the AM envelope.
- In successful modulation, this envelope is an exact replica of the original message signal $m(t)$.
- Proper envelope detection is possible only when the signal is not over-modulated ($\mu \leq 1$).

5.2.5 Block Diagram of AM Transmitter

A typical AM transmitter consists of the following main blocks:

- **Audio Amplifier:** Amplifies the weak audio signal from the microphone to a level suitable for modulation.
- **Carrier Oscillator:** Generates a stable high-frequency carrier signal at the desired transmission frequency.
- **Modulator:** The core component where the carrier amplitude is varied according to the modulating signal. This is typically implemented using a nonlinear device or circuit.
- **Power Amplifier:** Amplifies the modulated signal to the required power level for transmission.
- **Antenna:** Radiates the amplified AM signal into space for transmission to distant receivers.

The modulating signal and carrier signal are fed into the modulator circuit, which produces the amplitude-modulated output. The modulator is the key component that performs the multiplication of the two signals, either directly or through nonlinear processing.



5.2.6 Advantages of Amplitude Modulation

- **Simple Transmitter and Receiver Design:** AM transmitters and receivers have relatively simple circuit designs compared to other modulation schemes. AM uses basic circuits such as envelope detectors, making the system easy to design, understand, and maintain.
- **Low Cost of Implementation:** The simple hardware requirements make AM systems economical and widely affordable.

- **Wide Coverage Area:** AM signals can travel long distances using ground-wave and sky-wave propagation, especially in MF and HF bands.
- **Small Bandwidth:** Requires less bandwidth ($2f_m$) compared to FM.

5.2.7 Disadvantages of Amplitude Modulation

- **Highly Susceptible to Noise:** Noise (atmospheric, electrical) directly affects signal amplitude. Since information is encoded in amplitude, AM is highly susceptible to interference and static.
- **Poor Power Efficiency:** A large portion of transmitted power is wasted in the carrier and redundant sidebands, reducing overall efficiency.
- **Redundancy:** Both sidebands carry the same information, wasting spectrum.
- **Limited Fidelity:** Narrow bandwidth and noise susceptibility result in lower audio quality compared to FM/PM.

5.2.8 Applications of Amplitude Modulation

- **AM Radio Broadcasting:** Used in medium-wave and short-wave broadcasting for news, talk shows, and public announcements.
- **Aviation Communication:** AM is preferred for aircraft communication because overlapping signals can still be detected without complete signal loss.
- **Analog Television (Historical):** The video signal in traditional analog TV used a form of AM called Vestigial Sideband Modulation (VSB).
- **Simple Wireless Communication Systems:** Suitable where low cost and simplicity are more important than high audio quality.

5.3 Frequency Modulation (FM)

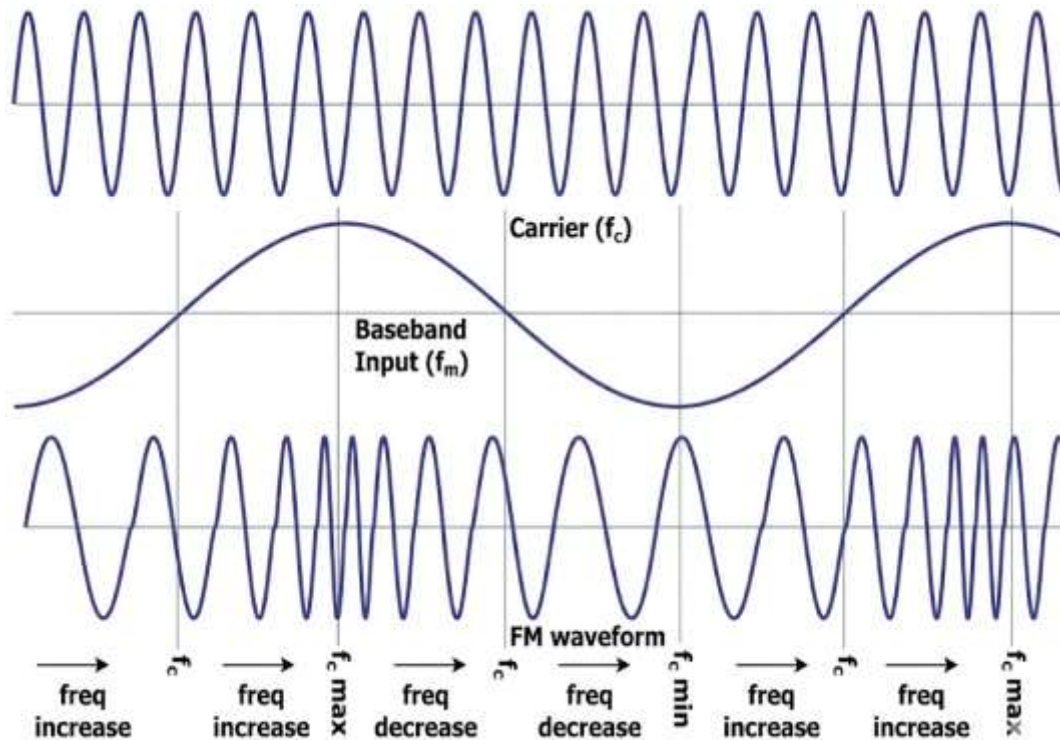
Frequency Modulation (FM) is a modulation technique in which the frequency of the carrier signal is varied in proportion to the instantaneous value of the modulating signal, while the amplitude of the carrier remains constant. In FM, information is conveyed through changes in the instantaneous frequency of the carrier wave, rather than through amplitude variations.

The basic principle of frequency modulation is that when a modulating signal is applied, the carrier frequency increases when the modulating signal amplitude is positive and decreases when it is negative. The amount of frequency change is

proportional to the amplitude of the modulating signal, while the rate of frequency change corresponds to the frequency of the modulating signal.

5.3.1 FM Waveforms

When the message signal voltage increases, the carrier frequency increases (waves become closer). When the message voltage decreases, the carrier frequency decreases (waves spread apart).



5.3.2 Frequency Deviation (Δf)

Frequency deviation (Δf) is defined as the maximum change in the instantaneous frequency of the carrier from its unmodulated value. It represents the extent to which the carrier frequency varies from its center frequency during modulation. Frequency deviation is directly proportional to the amplitude of the modulating signal.

Mathematically, frequency deviation is expressed as:

$$\Delta f = k_f \times A_m$$

where k_f is the frequency sensitivity constant of the modulator (in Hz/volt) and A_m is the amplitude of the modulating signal. The instantaneous frequency of the FM wave varies between $(f_c - \Delta f)$ and $(f_c + \Delta f)$.

Unit of Frequency Deviation: The unit of frequency deviation is Hertz (Hz). In commercial FM broadcasting, the maximum frequency deviation is standardized at ± 75 kHz from the carrier frequency. In narrowband FM systems used for voice communication, the deviation is typically limited to ± 5 kHz.

5.3.3 Modulation Index in FM

The modulation index of FM (β) is defined as the ratio of frequency deviation to the modulating frequency.

$$\beta = \Delta f / f_m$$

where,

Δf = maximum frequency deviation

f_m = modulating frequency

5.3.4 Significance of Modulation Index β

The modulation index determines the bandwidth occupied by the FM signal and the distribution of power among the carrier and sidebands. Unlike AM, the modulation index in FM can be greater than 1, and in wideband FM, it is typically much greater than 1. Larger β leads to better noise performance.

FM systems are classified based on the modulation index:

- **Narrowband FM (NBFM):** When $\beta \ll 1$, the system is called narrowband FM. NBFM has a bandwidth approximately equal to $2f_m$, similar to AM, and is used in applications like two-way mobile communication.
- **Wideband FM (WBFM):** When $\beta \gg 1$, the system is called wideband FM. WBFM occupies much greater bandwidth but provides superior noise immunity and signal quality. Commercial FM radio broadcasting uses wideband FM with β typically around 5.

5.3.5 Block Diagram of FM Transmitter

A typical FM transmitter consists of the following main blocks:

- **Pre-Amplifier:** The electrical signal generated by input transducer, such as a microphone, is extremely weak (usually in the millivolt range). The pre-amplifier boosts this weak voltage to a level sufficient to drive the next stage, which is the modulator. By amplifying the signal early, it ensures that the message signal remains significantly stronger than any internal circuit noise introduced in later stages thereby improving signal-to-noise ratio.

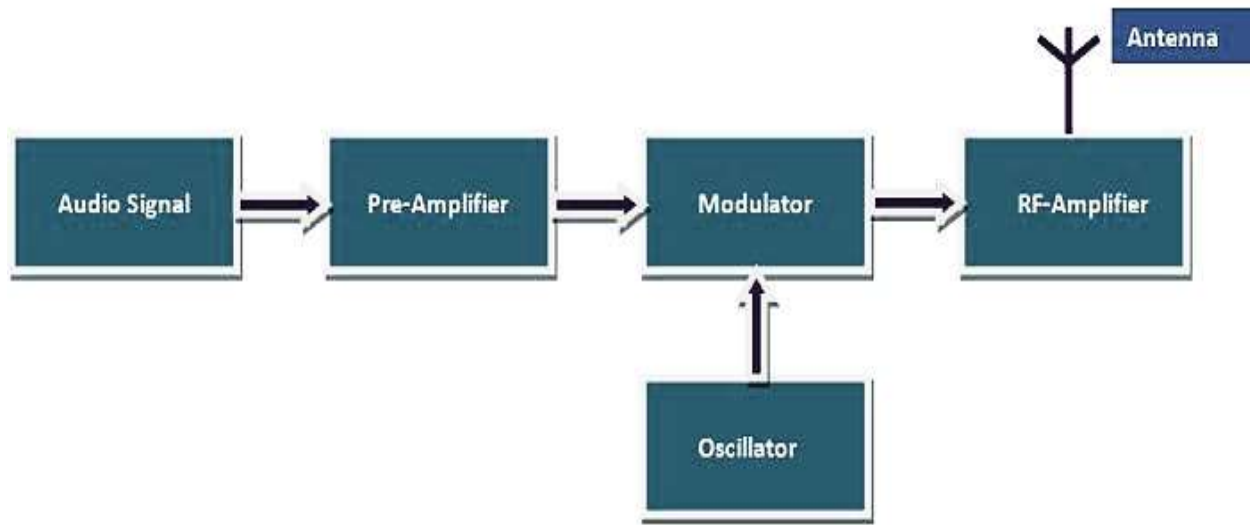


Fig. Block Diagram of FM Transmitter

- **RF Oscillator:** Generates a stable high-frequency carrier signal at the desired transmission frequency.
- **FM Modulator:** The actual "Frequency Modulation" process occurs by combining the message signal and the carrier wave. It superimposes the low-frequency electrical message signal (from the pre-amplifier) onto the high-frequency carrier wave. It generates the frequency-modulated signal using either direct or indirect method.
- **RF Power Amplifier:** While the pre-amplifier deals with voltage, the RF amplifier focuses on Power. To transmit a signal over long distances through free space, the carrier wave must have high power (measured in Watts) to overcome path loss and attenuation.
- **Antenna:** The Antenna is the final stage of the transmitter and acts as the interface between the electronic circuits and the physical environment. It acts as an output transducer, converting the high-frequency electrical current (power) into electromagnetic (EM) waves. Its primary function is to radiate these EM waves into free space (the communication channel) so they can propagate toward the receiver.

5.3.6 Advantages of Frequency Modulation

- **High Noise Immunity:** FM is highly resistant to noise and interference because noise primarily affects the amplitude of signals, while FM carries information in frequency variations. Hence, amplitude noise has minimal effect.

- **Better Audio Quality:** Wide bandwidth enables high-fidelity sound reproduction, ideal for music.
- **Constant Amplitude Signal:** Allows the use of efficient power amplifiers, improving transmitter efficiency.
- **Improved Signal-to-Noise Ratio:** The use of pre-emphasis and de-emphasis techniques in FM systems significantly improves the signal-to-noise ratio, particularly for high-frequency audio components.
- **Efficient Use of Transmitter Power:** All transmitted power in FM is useful, as the carrier amplitude remains constant and all power variations convey information, unlike AM where most power is in the non-information-bearing carrier.

5.3.7 Disadvantages of Frequency Modulation

- **Large Bandwidth Requirement:** FM signals, particularly wideband FM, require much greater bandwidth than AM. For example, commercial FM broadcasting uses approximately 200 kHz per channel compared to 10 kHz for AM, making FM less spectrum-efficient. Hence, limiting the number of available channels.
- **Complex Transmitter and Receiver Circuits:** FM modulators, discriminators, and AFC increase system complexity and cost.
- **Limited Coverage (Line-of-Sight):** FM signals, which are typically transmitted in the VHF and UHF bands, follow line-of-sight propagation and do not benefit from ionospheric reflection like AM. This limits the coverage area of FM stations to typically 50-100 km from the transmitter.
- **Requires More Infrastructure:** Due to limited range, FM broadcasting requires more transmitter stations and repeaters to cover the same geographic area as AM, increasing infrastructure costs.

5.3.8 Applications of Frequency Modulation

- FM Radio Broadcasting (88–108 MHz)
- Used for high-quality music and entertainment programs.
- Television Audio Transmission
- FM is employed due to its superior noise performance.

- Two-Way Radio Communication
- Used in walkie-talkies, police, emergency, and amateur radio systems.

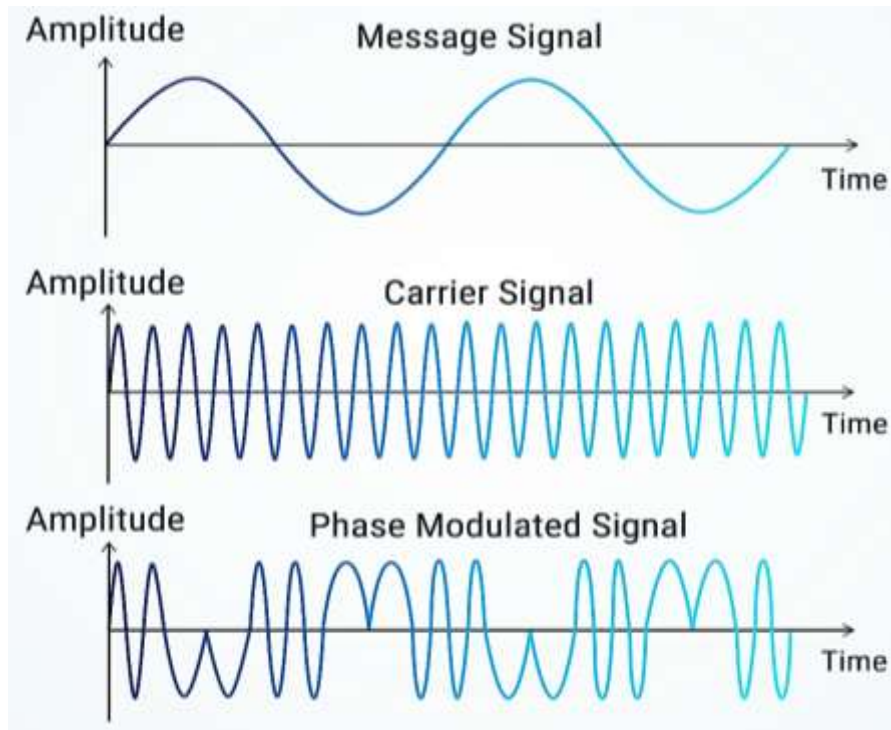
5.3.9 Comparison: AM vs. FM

Parameter	Amplitude Modulation (AM)	Frequency Modulation (FM)
Varied Parameter	Amplitude	Frequency
Carrier Amplitude	Varies with modulation	Constant
Modulation Index	$\mu = \frac{A_m}{A_c}$; $0 \leq \mu \leq 1$	$\beta = \frac{\Delta f}{f_m}$; $\beta \gg 1$ possible
Bandwidth	$BW = 2f_m$ (narrow, ~10 kHz)	$BW \approx 2(\Delta f + f_m)$ (wide, ~200 kHz)
Noise Immunity	Poor (noise affects amplitude directly)	Excellent (limiter removes amplitude noise)
Power Efficiency	Low (carrier power wasted)	High (constant envelope enables efficient PAs)
Fidelity	Low (audio BW ≤ 5 kHz)	High (audio BW up to 15 kHz)
Propagation	Long-range (ground/skywave in MF/HF)	Short-range (line-of-sight in VHF/UHF)
Circuit Complexity	Simple (envelope detector)	Complex (limiter, discriminator, AFC)
Typical Uses	AM radio, aviation communication	FM radio, TV audio, mobile communication

5.4 Phase Modulation (PM)

Phase Modulation (PM) is a modulation technique in which the phase of the carrier signal is varied in accordance with the instantaneous value of the modulating signal, while the amplitude and frequency of the carrier remain constant. In PM, information is conveyed through variations in the phase angle of the carrier wave.

The basic principle of phase modulation is that when a modulating signal is applied, the phase of the carrier signal changes instantaneously according to the amplitude of the modulating signal. A positive modulating signal causes a phase advance, while a negative modulating signal causes a phase delay relative to the unmodulated carrier.



5.4.1 Phase Deviation in PM

Phase deviation ($\Delta\phi$) is the maximum change in the carrier phase caused by the message signal.

- Higher message amplitude $\rightarrow$ larger phase deviation
- Lower message amplitude $\rightarrow$ smaller phase deviation

This phase variation carries the information.

5.4.2 Modulation Index in PM

The modulation index of PM is equal to the maximum phase deviation.

$$\beta_p = \Delta\phi = k_p A_m$$

where (A_m) is the peak amplitude of the message signal.

Important Observations

- In PM, modulation index depends only on message amplitude
- In FM, modulation index depends on frequency deviation and message frequency.
- PM modulation index is independent of message frequency.

5.4.3 Relationship Between FM and PM

Phase modulation and frequency modulation are closely related, as both fall under the category of angle modulation. In fact, phase modulation can be considered as a special case of frequency modulation, and vice versa. The key relationship is that phase is the integral of frequency, and frequency is the derivative of phase with respect to time.

When a signal is phase modulated, the instantaneous frequency also varies, making it appear similar to FM. Similarly, when a signal is frequency modulated, the phase also changes. The fundamental difference is:

- In FM, the instantaneous frequency deviation is directly proportional to the amplitude of the modulating signal.
- In PM, the phase deviation is directly proportional to the amplitude of the modulating signal, which means the instantaneous frequency deviation is proportional to the derivative of the modulating signal.

This relationship can be expressed mathematically: FM can be generated from PM by first integrating the modulating signal before applying it to a phase modulator. Conversely, PM can be generated from FM by first differentiating the modulating signal before applying it to a frequency modulator.

PM and FM are mathematically related angle modulation techniques.

- PM to FM: Integrate the message signal before phase modulation
- FM to PM: Differentiate the message signal before frequency modulation
- In FM: Frequency deviation $\propto$ message amplitude
- In PM: Frequency deviation $\propto$ derivative of message signal

5.4.4 Advantages of Phase Modulation (PM)

- **Good Noise Immunity:** PM provides excellent noise immunity because noise primarily affects amplitude, while information is carried in the phase of the signal.
- **Constant Amplitude Signal:** Like FM, PM maintains constant carrier amplitude allowing the use of efficient power amplifiers.
- **Better Frequency Stability:** PM systems can achieve better frequency stability than direct FM because the carrier frequency is generated by a stable crystal oscillator and only the phase is modulated.
- **Foundation for Digital Modulation:** PM is particularly well-suited for digital communication systems. Digital phase modulation techniques like

BPSK, QPSK, and QAM are highly efficient and form the basis of modern digital communication systems.

5.4.5 Disadvantages of Phase Modulation (PM)

- **Complex Receiver Design:** PM demodulation is more complex than AM demodulation. It requires sophisticated circuits or digital signal processing, particularly for coherent detection of digital PM schemes.
- **Requires Differentiation/Integration:** Depending on the system design, additional signal processing (differentiation or integration) may be required to properly recover the modulating signal from PM.
- **Limited Use in Analog Broadcasting:** PM is less common than FM for analog audio transmission.
- **Higher Bandwidth Requirement:** Like FM, PM typically requires wider bandwidth than AM, especially for wideband PM applications.

5.4.6 Applications of Phase Modulation (PM)

- Digital Communication Systems (PSK, QPSK)
- Satellite Communication due to better noise performance
- Telemetry and Data Transmission
- Phase-Locked Loops (PLL) and synchronization systems
- Indirect FM Generation

5.4.7 Comparison: AM vs FM vs PM

Parameter	AM	FM	PM
Varied Parameter	Amplitude of carrier signal	Frequency of carrier signal	Phase of carrier signal
Carrier Amplitude	Varies	Constant	Constant
Math Form	$A_c[1 + \mu m(t)]\cos(2\pi f_c t)$	$A_c\cos[2\pi f_c t + 2\pi k_f \int m dt]$	$A_c\cos[2\pi f_c t + k_p m(t)]$
Modulation Index	$\mu = \frac{A_m}{A_c}$ $0 \leq \mu \leq 1$	$\beta_f = \frac{\Delta f}{f_m}$ $\beta_f \gg 1$ possible	$\beta_p = k_p A_m$ (radians, no upper limit)
Freq. Deviation	Not applicable	$\Delta f = k_f A_m$	$\Delta f = \beta_p f_m = k_p f_m A_m$
Bandwidth	$2f_m$ (narrow, kHz) ~10	$2(\Delta f + f_m)$ (wide, ~200 kHz)	$2(\Delta f + f_m)$ (similar to FM)

Parameter	AM	FM	PM
Noise Immunity	Poor	Excellent	Very Good (needs coherent detection)
Power Efficiency	Low	High	High
Circuit Complexity	Simple	Moderate-Complex	Complex
Demodulation	Envelope detector	Discriminator, PLL	Coherent detection
Digital Equivalent	ASK	FSK	PSK
Typical Applications	AM radio, aviation	FM radio, TV audio	Digital comm, satellite

5.5 Introduction to Multiplexing

In a telecommunication system, several users need to transmit information simultaneously over a common communication medium — such as a coaxial cable, optical fiber, microwave link, or wireless channel. Providing a separate physical channel for each user would be both costly and impractical. Multiplexing is the technique that solves this problem by allowing multiple independent signals to share a single transmission medium efficiently.

5.5.1 Definition

Multiplexing is a technique by which multiple independent signals are combined into one composite signal, transmitted over a shared communication channel, and then separated back into the original individual signals at the receiving end.

5.5.2 Need for Multiplexing

Multiplexing is essential in modern communication for the following reasons:

- **Efficient Bandwidth Utilisation:** Most communication media possess a much higher bandwidth than a single user requires. Multiplexing allows this excess bandwidth to be shared among several users, ensuring no capacity is wasted.
- **Cost Reduction:** Installing separate cables or links for every user is expensive. A single multiplexed channel can serve many users, drastically reducing infrastructure and maintenance costs.
- **Increased System Capacity:** Multiplexing makes it possible to serve a large number of users over a limited number of physical links, increasing the overall capacity of the communication network.

- **Scalability:** Large networks — telephone systems, internet backbones, and satellite systems — depend on multiplexing to handle growing numbers of users without building entirely new infrastructure.

5.5.3 Key Terminologies

- **Multiplexer (MUX):** A multiplexer is a device that accepts multiple input signals from different sources and combines them into a single composite output signal for transmission over a shared communication channel. The multiplexer ensures that each input signal maintains its identity within the combined signal so it can be recovered at the destination.
- **Demultiplexer (DEMUX):** A demultiplexer performs the opposite function - it receives the composite multiplexed signal from the transmission channel and separates it back into the original individual signals, directing each to its appropriate destination. The demultiplexer must be synchronized with the multiplexer to correctly identify and extract each individual signal.
- **Channel Capacity:** Channel capacity is the maximum rate at which information can be reliably transmitted over a communication channel. It depends on the available bandwidth and the signal-to-noise ratio. Multiplexing is designed to utilise this capacity optimally by ensuring the channel carries multiple signals rather than lying partially idle.

5.5.4 Types of Multiplexing

Based on the manner in which the channel is shared, multiplexing is classified into two fundamental types:

- **Frequency Division Multiplexing (FDM):** The available bandwidth is divided into separate frequency bands; all signals are transmitted simultaneously, each occupying a different frequency band.
- **Time Division Multiplexing (TDM):** The entire channel bandwidth is shared in time; each signal is transmitted sequentially in its own dedicated time slot.

Other techniques such as Wavelength Division Multiplexing (WDM) — used in optical fiber — and Code Division Multiplexing (CDM) — used in cellular networks — also exist, but FDM and TDM are the foundational methods.

5.6 Frequency Division Multiplexing (FDM)

5.6.1 Definition

Frequency Division Multiplexing (FDM) is an analog multiplexing technique in which the total available bandwidth of a communication channel is divided into several non-overlapping frequency sub-bands. Each input signal is assigned one sub-band and all signals are transmitted simultaneously, each occupying a different frequency range. Separation of signals is therefore achieved in the frequency domain.

5.6.2 Working Principle

The FDM process takes place in two stages — multiplexing at the transmitter and demultiplexing at the receiver.

Transmitter (Multiplexing)

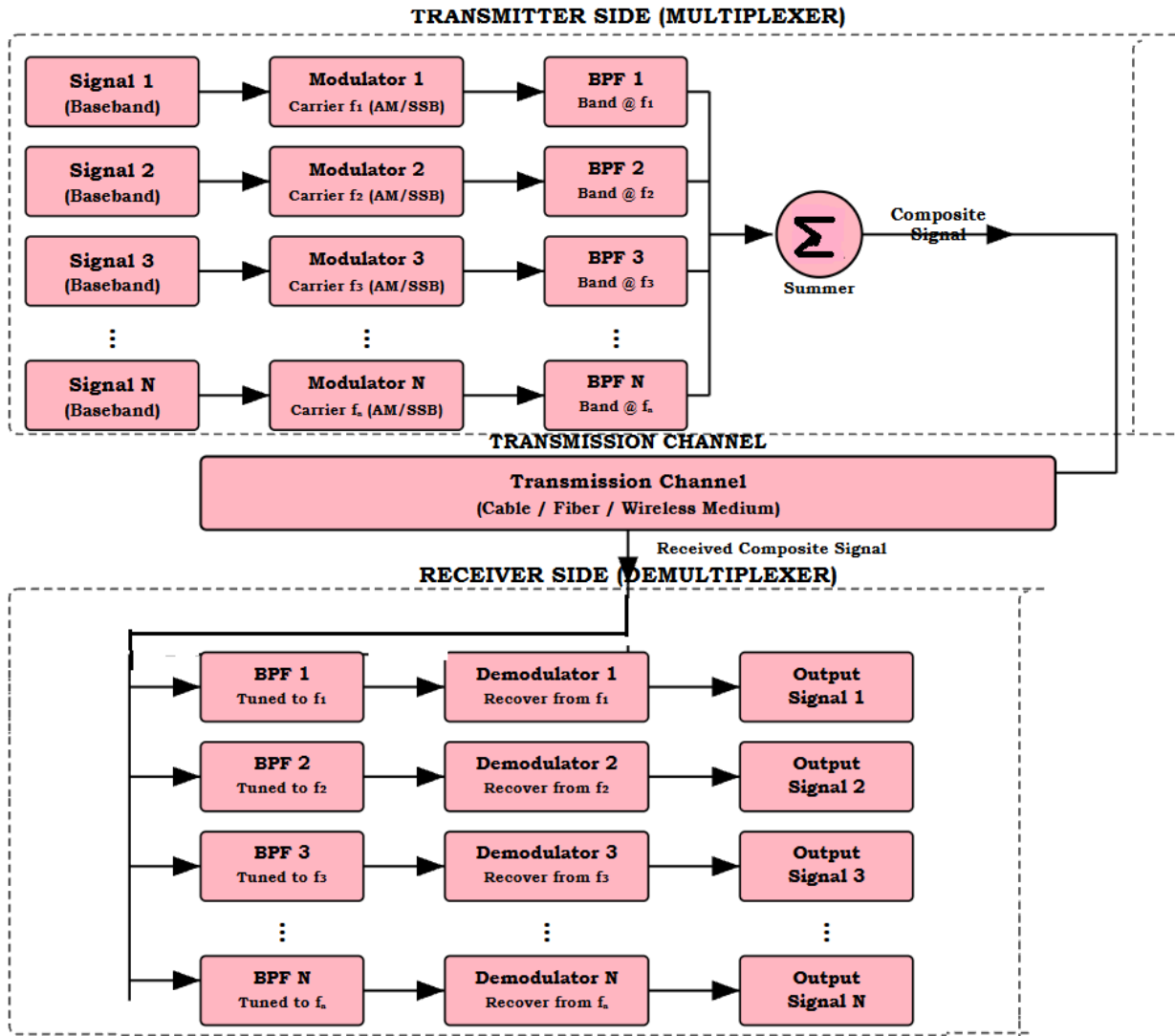
Each input signal — which exists at baseband frequency — is fed into a separate modulator. Each modulator uses a different sub-carrier frequency ($f_1, f_2, f_3, \dots$) so that the resulting modulated signals occupy distinct, non-overlapping frequency bands. The modulated signals are then passed through individual bandpass filters to confine each signal strictly within its allocated band. All filtered signals are finally combined by a summing network to form a single composite signal, which is transmitted over the shared communication channel.

Receiver (Demultiplexing)

At the receiver, the composite signal is applied simultaneously to a bank of bandpass filters, each tuned to one specific sub-carrier frequency. Each filter extracts only the signal in its allocated band and rejects all others. The extracted signal is then passed through a demodulator, which removes the sub-carrier and recovers the original baseband signal. The recovered signals are delivered to their respective destinations.

5.6.3 Block Diagram of an FDM System (Pending)

Figure shows the complete FDM system. On the transmit side, N input signals are modulated onto N different sub-carriers and combined. On the receive side, N bandpass filters separate the channels before demodulation.



Transmitter Side (Multiplexer):

A typical FDM transmitter consists of the following main components:

- **Input Signals:** Multiple independent baseband signals (Signal 1, Signal 2, Signal 3, etc.) that need to be transmitted simultaneously.
- **Modulators:** Each input signal is fed into a separate modulator. Each modulator shifts its input signal to a different carrier frequency (f_1 , f_2 , f_3 , etc.). Common modulation techniques used include Amplitude Modulation (AM) or Single Sideband modulation (SSB).
- **Bandpass Filters:** After modulation, each signal passes through a bandpass filter that ensures the signal stays within its allocated frequency band and removes any unwanted frequency components.

- **Combining Network (Summer):** All the filtered, modulated signals are combined in a linear mixer or summing circuit to create a single composite signal.
- **Transmission Channel:** The composite signal is transmitted through the communication channel (cable, fiber, or wireless medium).

Receiver Side (Demultiplexer):

The FDM receiver consists of the following components:

- **Received Composite Signal:** The multiplexed signal received from the channel contains all the individual signals combined together.
- **Bank of Bandpass Filters:** The received signal is fed into multiple bandpass filters arranged in parallel. Each filter is tuned to extract one specific frequency band corresponding to one of the original signals.
- **Demodulators:** After filtering, each extracted signal is fed into a demodulator that recovers the original baseband signal from the carrier frequency.
- **Output Signals:** The demodulated signals are delivered to their respective destinations, completely separated and independent once again.

5.6.4 Guard Bands

When multiple modulated signals are placed adjacent to each other in frequency, there is a risk of overlap due to imperfect filters and carrier frequency drift. To prevent this, small unused frequency gaps called guard bands are deliberately inserted between adjacent channels.

Purpose of Guard Bands

- **Prevention of crosstalk:** Adjacent signals may have sidebands that spill into neighboring bands. Guard bands provide the necessary frequency separation to prevent this overlap.
- **Accommodation of filter roll-off:** Practical bandpass filters do not have a perfectly sharp cutoff; their response rolls off gradually. Guard bands ensure the transition region of one filter does not fall within the signal band of the next channel.
- **Tolerance for frequency drift:** Sub-carrier oscillators may drift slightly due to temperature or aging. Guard bands absorb such drift without causing interference.

The width of a guard band is a design trade-off: wider guard bands give better isolation but reduce the number of channels that can fit in the available bandwidth. Typical guard bands range from a few hundred hertz to several kilohertz.

5.6.5 Crosstalk in FDM

Crosstalk is unwanted interference in which signal energy from one channel leaks into an adjacent channel. In FDM, it arises mainly from inadequate frequency separation, imperfect filters, or non-linear amplifier behavior (intermodulation distortion). Crosstalk is minimised by providing adequate guard bands, using high-quality sharp-cutoff filters, and careful frequency planning during system design.

5.6.6 Advantages of FDM

- **Simultaneous transmission:** All channels transmit continuously at the same time, making FDM ideal for real-time services such as radio and television broadcasting.
- **No synchronisation required:** Each channel operates independently at its assigned frequency; precise timing coordination between transmitter and receiver is not needed.
- **Simple demultiplexing:** Separating channels using bandpass filters is straightforward and well understood.
- **Versatility:** FDM can carry both analog and digital signals.

5.6.7 Disadvantages of FDM

- **Bandwidth wastage:** Guard bands represent unused spectrum. In systems with many channels, the cumulative waste can be significant.
- **Crosstalk and intermodulation:** Non-ideal filters and non-linear components can cause interference between channels.
- **Hardware complexity:** Each channel requires its own modulator, demodulator, and bandpass filter, increasing cost and power consumption.
- **Fixed channel allocation:** Once frequency bands are assigned, changing them requires hardware modifications.

5.6.8 Applications of FDM

- **AM and FM radio broadcasting:** Multiple radio stations transmit simultaneously in a city, each occupying a different frequency band.

- **Television broadcasting:** Terrestrial TV channels use FDM; each channel occupies a fixed bandwidth slot (e.g., 6 MHz per channel in many standards).
- **Cable television (CATV):** Hundreds of channels are delivered over a single coaxial cable, each in its own frequency slot.
- **Satellite communication:** Multiple uplink and downlink channels are multiplexed onto a single satellite transponder.
- **First-generation (1G) cellular systems:** The AMPS system used FDM to divide the cellular spectrum into individual 30 kHz voice channels.

5.7 Time Division Multiplexing (TDM)

Time Division Multiplexing (TDM) is a digital multiplexing technique in which multiple signals share the entire bandwidth of a communication channel, but each signal is transmitted during a different, dedicated time slot. At any given instant, only one signal uses the channel. Separation of signals is therefore achieved in the time domain.

5.7.1 Working Principle

In TDM, each input signal is sampled periodically at a rate satisfying the Nyquist criterion (at least twice the highest signal frequency). The samples from all channels are interleaved in time — one sample from Channel 1, followed by one from Channel 2, then Channel 3, and so on. A high-speed electronic switch (commutator) at the transmitter connects each input to the channel in turn for a brief time slot. After one complete round of all channels, the cycle repeats, forming a continuous stream of interleaved samples.

At the receiver, a synchronised switch (de-commutator) routes each incoming sample to its correct output. The output samples are held and then smoothed by low-pass reconstruction filters to recover the original continuous signals. The receiver switch must operate in perfect synchrony with the transmitter switch; this synchronisation is the most critical requirement of TDM.

5.7.2 Time Slots and Their Significance

A time slot is the fixed time interval allocated to each signal during which it has exclusive access to the transmission channel. The duration of each time slot must be sufficient to transmit one complete sample of the signal. The collection of time slots for all signals, transmitted in sequence, forms one complete cycle called a frame.

Significance of Time Slots

- **Deterministic Access:** Each signal is guaranteed access to the channel at predictable intervals, ensuring reliable transmission with bounded delay.
- **Bandwidth Sharing:** Time slots enable multiple signals to share the full channel bandwidth fairly, with each getting a fraction of the total transmission time proportional to its needs.
- **Quality of Service:** The time slot allocation can be adjusted to provide different levels of service to different signals. Critical signals can be assigned more time slots per frame, ensuring higher data rates or better quality.
- **Synchronization Reference:** Time slots provide a timing structure that helps maintain synchronization between transmitter and receiver. The regular pattern of time slots serves as a clock reference.

5.7.3 Frame Structure in TDM

In TDM systems, the term 'frame' refers to one complete cycle of time slots for all channels. A frame consists of multiple time slots arranged sequentially, with each slot dedicated to one specific channel. The frame repeats continuously, allowing each signal to be sampled and transmitted at regular intervals.

Typical Frame Structure Components

- **Synchronization Bits:** Each frame begins with a unique bit pattern that identifies the start of the frame. This helps the receiver maintain synchronization and know when each channel's time slot begins.
- **Channel Time Slots:** Following the sync bits are the time slots for each channel. For example, in a system with 8 channels, there would be 8 time slots per frame (Slot 1, Slot 2, ... Slot 8).
- **Control Information:** Some frames may include additional time slots for control signals, alarm indicators, or maintenance data.
- **Framing Overhead:** The synchronization and control bits represent overhead - they do not carry user data but are essential for system operation.

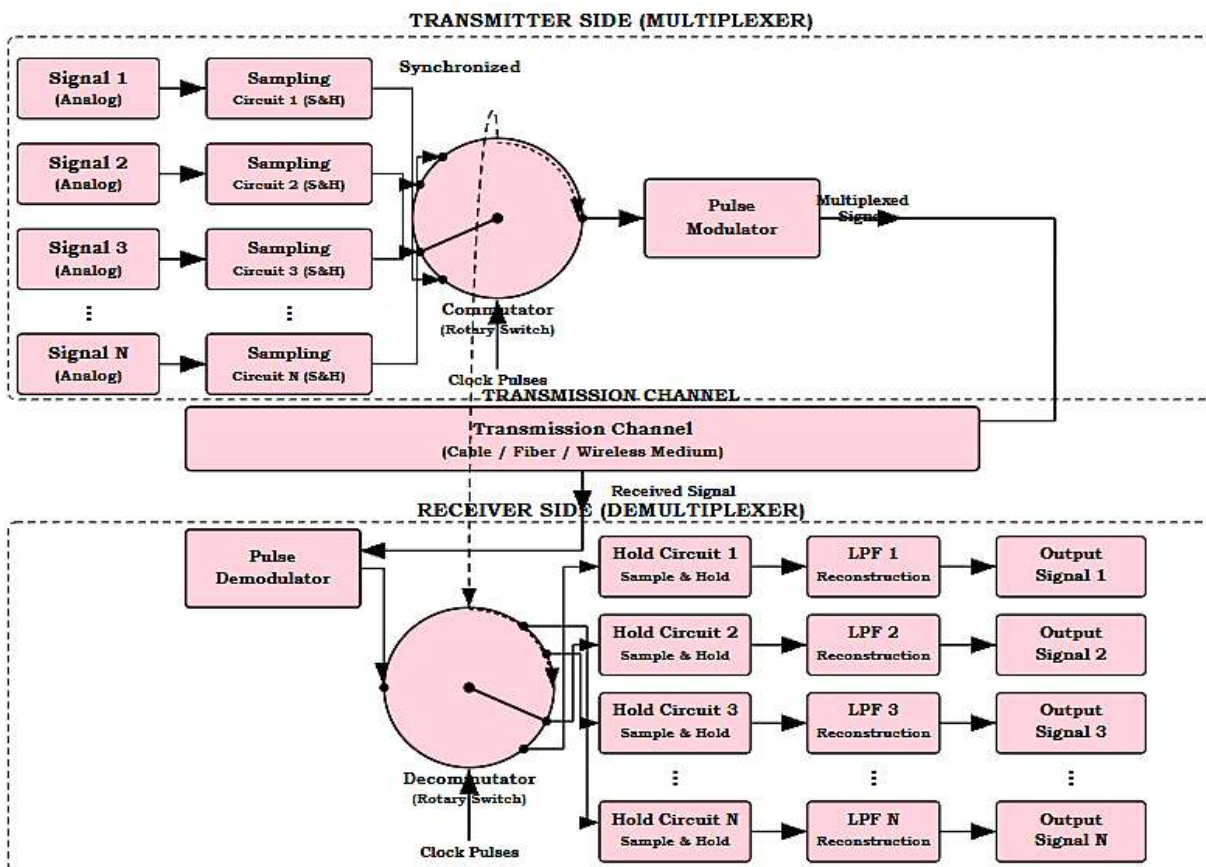
Example, in digital telephone systems (PCM), each voice channel is sampled at 8000 samples per second, so the frame rate is 8000 frames per second, with each frame lasting 125 microseconds.

5.7.4 Block Diagram of a TDM System

Transmitter Side (Multiplexer):

A typical TDM transmitter consists of:

- **Input Signals:** Multiple independent signals (typically analog) that need to be multiplexed.
- **Sampling Circuits:** Each input signal passes through a sample-and-hold circuit that captures instantaneous signal values at regular intervals.
- **Time Division Multiplexer (Electronic Switch – Commutator):** A high-speed rotary or electronic switch that sequentially connects each sampled signal to the output line. The switch cycles through all inputs rapidly, giving each input a brief time slot.
- **Timing and Control Unit:** Generates clock signals that control the switching sequence and timing. This ensures each channel gets its allocated time slot in the correct order.
- **Transmission Channel:** The multiplexed signal (interleaved samples) is transmitted through the communication channel.



Receiver Side (Demultiplexer):

The TDM receiver consists of:

- **Received Multiplexed Signal:** The time-interleaved samples arrive from the transmission channel.
- **Synchronization Circuit:** Detects the synchronization pattern in the received frames to maintain proper timing alignment with the transmitter.
- **Time Division Demultiplexer (Electronic Switch – De-commutator):** A synchronized switch that routes each received sample to its corresponding output channel. The switch must operate in perfect synchrony with the transmitter's multiplexer.
- **Hold Circuits:** Each output has a sample-and-hold circuit that retains each sample value until the next sample arrives.
- **Low-Pass Filters:** Reconstruction filters smooth the sampled signals back into continuous waveforms, recovering the original analog signals.

5.7.5 Synchronisation in TDM

Synchronisation is the process of aligning the timing of the receiver's de-commutator with the transmitter's commutator so that each incoming sample is routed to the correct output channel. It is the most critical requirement of a TDM system.

Why Synchronisation is Critical?

- **Correct channel assignment:** Without synchronisation, the receiver does not know which time slot belongs to which channel, and samples are delivered to wrong outputs.
- **Frame alignment:** The receiver must identify the start of every frame to interpret the slot sequence correctly.
- **Timing accuracy:** Even small clock differences accumulate over successive frames, leading to progressive misalignment and eventual loss of all channels simultaneously.

Consequences of Lost Synchronisation

Loss of synchronisation causes all channels to fail simultaneously. Samples from one channel appear at another channel's output (channel mixing), making all received data unusable. Service is fully interrupted until synchronisation is re-established, which may take several frames.

5.7.6 Types of TDM

(a) Synchronous TDM

In synchronous TDM, each channel is assigned a fixed time slot in every frame, regardless of whether the channel has data to transmit. If a channel is idle, its slot is transmitted empty. This approach is simple and predictable, but wastes bandwidth when channels are not continuously active. Synchronous TDM is used in digital telephony (T1/E1 systems) where every voice channel is sampled at a constant, known rate.

(b) Asynchronous (Statistical) TDM

In asynchronous (statistical) TDM, time slots are allocated dynamically — only to channels that actually have data to send at a given moment. Idle channels receive no slot allocation, so the available bandwidth is distributed entirely among active channels. Each transmitted slot must carry an address or channel identifier so the receiver knows which channel the data belongs to. This overhead is small compared with the gain in bandwidth efficiency. Statistical TDM is widely used in computer networks and packet-switching systems where traffic is inherently bursty.

5.7.7 Advantages of TDM

- **Full bandwidth per channel:** During its time slot, a channel uses the entire channel bandwidth, maximising throughput.
- **No guard bands:** Unlike FDM, TDM requires no unused frequency gaps; only a small synchronisation overhead is needed.
- **No frequency-domain crosstalk:** Because only one channel is active at any instant, there is no simultaneous frequency overlap and hence no inter-channel interference.
- **Suited to digital systems:** TDM works naturally with digital signals and is easily implemented with high-speed digital switching circuits.
- **Flexible slot allocation:** In statistical TDM, slots are dynamically assigned, improving efficiency for bursty traffic.

5.7.8 Disadvantages of TDM

- **Precise synchronisation required:** Maintaining exact timing alignment between transmitter and receiver is complex and adds to system cost.
- **High-speed switching:** As the number of channels increases, the commutator and de-commutator must switch proportionally faster.
- **Propagation delay sensitivity:** Long links introduce delays that can make clock recovery and synchronisation harder to maintain.

- **Slot waiting delay:** A channel must wait for its assigned time slot before transmitting, introducing a small but deterministic delay.
- **Bandwidth waste in synchronous TDM:** Idle channels in synchronous TDM still occupy their slots, wasting bandwidth.

5.7.9 Applications of TDM

- **Digital telephony (T1/E1/PCM):** The T1 carrier (North America) multiplexes 24 voice channels using TDM in a 193-bit frame at 1.544 Mbps. The E1 carrier (Europe) carries 30 voice channels at 2.048 Mbps.
- **GSM cellular systems:** GSM uses a combination of FDM and TDM (TDMA). Within each 200 kHz frequency channel, time is divided into 8 slots serving up to 8 users.
- **Satellite communication:** Multiple earth stations share a satellite transponder using TDMA (Time Division Multiple Access), where each station transmits only during its allocated burst time slot.
- **Digital broadcasting (DAB/DVB):** Digital audio and video broadcasting systems use TDM to multiplex multiple program streams into a single transport stream.
- **Optical fibre networks (SONET/SDH):** Synchronous optical networks use high-speed TDM to aggregate many lower-rate tributaries onto a single fiber.

5.7.10 Comparison: FDM Vs TDM

Parameter	FDM	TDM
Basic Principle	Divides bandwidth into frequency bands	Divides time into slots
Bandwidth usage	Guard bands required	No guard bands
Signal type	Mainly analog	Mainly digital
Transmission	Simultaneous	Sequential
Synchronization	Not critical	Essential
Delay	No waiting delay	Time slot delay
Bandwidth efficiency	Lower	Higher
Flexibility	Fixed allocation	Dynamic allocation possible
Complexity	Simpler	More complex
Modulation Required	Yes (for each channel)	No (baseband transmission is possible)
Typical Applications	Radio/TV broadcasting, CATV	Digital telephony, computer networks

5.8 Error Control in Digital Communication

In any digital communication system, data is transmitted from a sender to a receiver through a physical medium such as copper wire, optical fiber, or wireless radio waves. During this transmission, the signal is exposed to various kinds of disturbances — collectively called noise — that can alter the original bit pattern. A bit that was sent as 1 may be received as 0, or a group of bits may be corrupted entirely. This phenomenon is known as a transmission error, and if left undetected or uncorrected, it leads to unreliable communication.

To address this fundamental challenge, engineers use a set of techniques collectively called error control. Error control involves two closely related but distinct operations: error detection and error correction.

Error Detection is the ability of the receiver to identify that one or more bits in the received data frame have been corrupted during transmission. The receiver does not attempt to fix the corrupted data on its own; it simply knows that an error has occurred and may request retransmission.

Error Correction, on the other hand, is the ability of the receiver to not only detect the error but also determine exactly which bit or bits are wrong, and to restore the original data without any retransmission. This is a far more powerful but also more complex capability.

The key difference between the two lies in the amount of additional information (redundancy) that must be added to the data. Error detection requires fewer extra bits, while error correction requires significantly more extra bits so that the receiver has enough information to reconstruct the original message.

5.8.1 Why Is Error Control Necessary?

The need for error control arises from the imperfect nature of every physical communication channel. Several real-world factors introduce errors into transmitted data.

Thermal Noise: Random electron movement in conductors generates a background level of electrical interference that is always present, especially at high data rates.

Impulse Noise: Sudden spikes of electrical energy — caused by lightning, switching of heavy electrical equipment, or faulty connectors — can corrupt entire sequences of bits at once.

Attenuation: As a signal travels over distance, its amplitude decreases. A weakened signal is more susceptible to noise and may be misinterpreted at the receiver.

Interference: In wireless systems, signals from other transmitters, mobile devices, or microwave ovens can overlap with the desired signal and cause bit corruption.

Cross-talk: In cable bundles, the electromagnetic field from one wire can induce unwanted signals in an adjacent wire, corrupting the data it carries.

Because these disturbances are unavoidable in practice, error control is an essential layer of protection in every digital communication system — from USB drives and Wi-Fi networks to satellite links and 5G cellular systems.

5.8.2 Types of Errors in Digital Communication

Errors in digital transmission are broadly classified into two categories based on the number of bits affected and their pattern.

Single-Bit Error

A single-bit error occurs when exactly one bit in a data unit is changed from 0 to 1, or from 1 to 0. All other bits in the frame remain intact. This type of error is more common in parallel data transmission, where individual bits travel on separate wires and only one wire may be affected by a disturbance at any given instant.

Sent: 0 1 0 0 **1** 1 0 1

Received: 0 1 0 0 **0** 1 0 1
 ^

Single-bit error at position 5 (1 flipped to 0)

Single-bit errors are relatively straightforward to detect and correct because only a small, isolated change has occurred in the data.

Burst Error

A burst error occurs when two or more consecutive bits in a data unit are corrupted. The length of a burst error is measured from the first corrupted bit to the last corrupted bit, and every bit within that span is considered part of the burst — though not all bits in that span need to be changed.

Sent: 0 1 0 1 0 1 1 0 1 0

Received: 0 1 1 0 1 0 1 0 1 0
 ^ ^ ^ ^

Burst of length 4 (positions 3 to 6 affected)

Burst errors are far more common in serial data transmission and in wireless systems, where a single event such as a lightning strike or multipath fading can

corrupt an entire sequence of bits passing through the channel at that moment. Burst errors are much harder to detect and correct than single-bit errors, and techniques such as CRC and interleaving are specifically designed to deal with them.

Parameter	Single-Bit Error	Burst Error
Bits affected	Exactly one bit	Two or more consecutive bits
Common cause	White / thermal noise	Impulse noise, fading, interference
Transmission type	Parallel transmission	Serial transmission
Difficulty to handle	Easier to detect & correct	Harder to detect & correct
Example technique	Parity check	CRC, LRC, Interleaving

5.8.3 Redundancy in Error Control Coding

The foundation of all error control schemes is the concept of redundancy. In the context of error control, redundancy means deliberately adding extra bits to the original data before transmission. These extra bits, called redundant bits or check bits, carry no new information about the message itself; their sole purpose is to enable the receiver to check whether the received data matches what was originally sent.

Consider a simple analogy: when you repeat a spoken word twice in a noisy environment, the listener has a second copy to cross-check. If one copy is heard incorrectly, the repetition helps identify or even recover the correct word. Redundant bits work on the same principle, but use mathematical relationships rather than simple repetition.

At the sender's side, the redundant bits are computed from the original data using a predefined mathematical rule or algorithm. These bits are then appended to or interspersed within the original data, and the complete frame (data bits + redundant bits) is transmitted.

At the receiver's side, the same algorithm is applied to the received frame. If the computed result matches what is expected, the data is considered error-free. If there is a mismatch, an error has been detected — and, depending on the technique used, the receiver may also be able to pinpoint and correct the error.

The trade-off is bandwidth efficiency: more redundant bits improve error-handling capability but reduce the proportion of the channel capacity used for actual data. Designers must balance these needs depending on the application.

5.9 Error Detection Techniques

Several well-established techniques are used for error detection in digital communication. The three most widely used are *parity checking, checksum, and cyclic redundancy check (CRC)*. Each offers a different level of error-detecting capability and computational complexity.

5.9.1 Parity Checking or Vertical Redundancy Check (VRC)

Parity checking is the simplest and oldest error detection method. It works by appending a single extra bit — called the parity bit — to a block of data bits such that the total number of 1s in the resulting codeword (data bits + parity bit) is either always even or always odd.

At the Sender's End

Even Parity

In even parity, the parity bit is chosen so that the total count of 1s in the codeword is even. If the original data already contains an even number of 1s, the parity bit is set to 0. If the data contains an odd number of 1s, the parity bit is set to 1.

Data byte: 1 0 1 1 0 0 1 0
Count of 1s: 4 (already even)
Parity bit: 0
Transmitted codeword: 1 0 1 1 0 0 1 0 **0**

Another example

Data byte: 1 0 0 0 0 1 1 0
Count of 1s: 3 (odd → needs 1 to be even)
Parity bit: 1
Transmitted codeword: 1 0 0 0 0 1 1 0 **1**

Odd Parity

In odd parity, the parity bit is chosen so that the total count of 1s in the codeword is always odd. If the original data already has an odd count of 1s, the parity bit is 0; otherwise, it is 1.

Data byte: 1 0 1 1 0 0 1 0
Count of 1s: 4 (even → needs to be odd)
Parity bit: 1
Transmitted codeword: 1 0 1 1 0 0 1 0 **1**

At the receiver's end, the count of 1s in the received codeword is recalculated. If the agreed-upon parity is even but the count is odd (or vice versa), an error is flagged. This type of check — performed on each row of data independently — is called Vertical Redundancy Check (VRC).

At the Receiver's End

When the receiver gets the codeword, it counts the total number of 1s (including the parity bit) and checks whether the result matches the agreed-upon parity scheme.

Suppose the sender transmitted the codeword using even parity:

Case 1 – No error detected (Even Parity)

Received codeword: 1 0 1 1 0 0 1 0 0
Count of 1s: 4 (even) ✓
Result: Parity check PASSED — no error detected

Case 2 – Error detected (Even Parity)

Suppose a bit was flipped during transmission (the 3rd bit changed from 1 → 0):

Transmitted codeword: 1 0 **1** 1 0 0 1 0 0 (4 ones — even)
Received codeword: 1 0 **0** 1 0 0 1 0 0 (3 ones — odd)
Count of 1s: 3 (odd) ✗
Result: Parity check FAILED — error detected

Since the receiver expected an even count but got an odd count, it flags an error and typically requests retransmission.

Case 3 – Error undetected (VRC limitation)

If two bits are flipped during transmission, the parity count remains even, and the error goes unnoticed:

Transmitted codeword: 1 0 **1 1** 0 0 1 0 0 (4 ones — even)
Received codeword: 1 0 **0 0** 0 0 1 0 0 (2 ones — even)
Count of 1s: 2 (even) ✓
Result: Parity check PASSED — but TWO errors actually occurred!

This is a key limitation of VRC: it can detect only an odd number of bit errors. Any even number of simultaneous bit flips will cancel out and go undetected.

Advantages of Parity Check (VRC)

- Simple and easy to implement
- Requires minimum additional hardware
- Suitable for basic error detection

Limitation of Parity Check (VRC)

The major limitation of single-parity (VRC) is that it can detect only an odd number of bit errors. If exactly two bits in a codeword are flipped, the parity count remains the same and the error goes undetected. To overcome this, a two-dimensional approach is used.

Applications

- Used in digital communication systems
- Employed in computer memory and data transmission
- Used as a basic error detection method in networks

5.9.2 Longitudinal Redundancy Check (LRC) — Two-Dimensional Parity

LRC, also known as two-dimensional parity, arranges the data bits in a two-dimensional grid (rows and columns). A parity bit is computed for each row (VRC) and also for each column. The column parity bits form an additional parity byte that is transmitted along with the data.

Data organised as 4 rows of 8 bits each (even parity):

Row 1: 1 0 1 1 0 0 1 0 → Parity bit: 0

Row 2: 0 1 1 1 0 1 1 0 → Parity bit: 1

Row 3: 1 1 0 1 1 0 1 0 → Parity bit: 1

Row 4: 1 0 0 0 1 1 1 0 → Parity bit: 0

Col P: 1 0 0 1 0 0 0 0 (column parity byte)

LRC significantly improves burst error detection. A single-bit burst error that corrupts one bit in a row will affect both the row parity (VRC) and the column parity (LRC), making it detectable. Even some patterns of two-bit errors that VRC

alone would miss can be caught by examining both the row and column parity bits together. LRC is therefore much more reliable for burst-prone channels.

At the Receiver's End

The receiver rebuilds the same 2D grid, recomputes row and column parities, and compares them with the received parity values.

Case 1 – No error

All recomputed row and column parities match the received ones → CHECK PASSED.

Case 2 – Single-bit error

Suppose bit 3 of Row 2 is flipped during transmission:

```

Row 1:  1 0 1 1 0 0 1 0 | 0 ✓
Row 2:  0 1 0 1 0 1 1 0 | 1 ✗ ← row parity mismatch
Row 3:  1 1 0 1 1 0 1 0 | 1 ✓
Row 4:  1 0 0 0 1 1 1 0 | 0 ✓

```

```

Col P:  1 0 0 1 0 0 0 0
          ✗
        col 3 mismatch

```

Row 2 and Col 3 both fail → error pinpointed at Row 2, Col 3. This is LRC's key advantage: it can locate the exact bit, making single-bit error correction possible.

Case 3 – Two-bit error

Suppose bits at Row 2 Col 3 and Row 2 Col 5 are both flipped:

```

Row 1:  1 0 1 1 0 0 1 0 | 0 ✓
Row 2:  0 1 0 1 1 1 1 0 | 1 ✓
Row 3:  1 1 0 1 1 0 1 0 | 1 ✓
Row 4:  1 0 0 0 1 1 1 0 | 0 ✓

```

```

Col P  1 0 0 1 0 0 0 0
          ✗  ✗
        Col3 Col5

```

Result: Error detected! Both flipped bits fall in different columns, so each column parity catches one of them independently. VRC alone would have missed this since two flips in the same row cancel each other out.

5.9.3 Checksum Method

The checksum method is widely used at the network and transport layers of communication protocols (for example, in IP, UDP, and TCP). It works by dividing the data into equal-sized segments and summing them up arithmetically. The result, after some processing, is appended to the data as the checksum value.

Checksum at the Sender

The sender divides the data into fixed-size blocks, typically 16 bits each. These blocks are added together using binary addition. If the addition produces a carry out of the most significant bit, that carry is wrapped around and added to the least significant bit — a process called end-around carry. The final sum is then complemented (all 0s become 1s and all 1s become 0s) to produce the checksum. This checksum is appended to the data and the complete unit is transmitted.

Data blocks (16-bit each):
Block 1: 1001 0110 1010 1011
Block 2: 0101 0011 0001 1111

Sum: 1110 1001 1100 1010
(no carry in this example)

Checksum = 1's complement of sum
= 0001 0110 0011 0101

Transmitted: Block 1 + Block 2 + Checksum

Checksum at the Receiver

The receiver adds all the received segments together, including the checksum itself. If the transmission was error-free, the sum of all segments (including the checksum) will produce a result in which all bits are 1 (i.e., 1111...1 in binary). Taking the 1's complement of this result gives all zeros. If the final result is all zeros, the data is accepted as correct. If any bit is not 1, an error has been detected.

Receiver adds: Block 1 + Block 2 + Checksum
Expected result: 1111 1111 1111 1111 (all ones)
1's complement: 0000 0000 0000 0000 (all zeros)

If result = 0000...0 → No error (data accepted)
If result ≠ 0000...0 → Error detected (data rejected)

The checksum method is fast, simple to implement in software, and effective for detecting most random errors. However, it has a weakness: if errors in different

blocks cancel each other out numerically, the incorrect data may still produce the correct checksum. CRC, described next, is far more robust in this regard.

Advantages

- More reliable than parity check
- Suitable for large data blocks

Limitation

- If errors in different blocks cancel each other out numerically, the incorrect data may still produce the correct checksum

Applications

- IP, TCP, UDP protocols
- File transfer systems

5.9.4 Cyclic Redundancy Check (CRC)

Cyclic Redundancy Check (CRC) is a highly powerful and widely used error-detection technique in digital communication and storage systems. Unlike simple parity checks, CRC uses polynomial division to verify data integrity, making it extremely effective at detecting common transmission errors, such as burst errors (multiple consecutive bits being corrupted).

Basic Working Principle

In CRC, the sender treats the data bits as a binary number and divides it by a predetermined binary value called the generator polynomial. The remainder obtained from this division is called the CRC code or CRC bits. These CRC bits are appended to the original data before transmission.

CRC at the sender:

- *Dataword*: The original binary data to be sent.
- *Divisor (Generator Polynomial)*: A fixed binary number agreed upon by both transmitter and receiver.
- *Step 1: Append Zeros to the Dataword*: The number of zeros to append is equal to $(n - 1)$, where n is the number of bits in the divisor.
- *Step 2: Binary Division (Modulo-2)*: In modulo-2 division, we use the XOR operation instead of subtraction.
 - $0 \oplus 0 = 0$
 - $0 \oplus 1 = 1$

- $1 \oplus 0 = 1$
- $1 \oplus 1 = 0$
- *Step 3: Form the Codeword: Remainder (CRC Bits):* The remainder of the division (from step 2) is appended to the dataword to form the codeword which is then transmitted. The quotient is discarded.

CRC at the receiver:

At the receiver side, the received data (including CRC bits) is again divided by the same generator polynomial. The remainder of this division is called the syndrome.

- If the syndrome is zero, the data is assumed to be error-free. The dataword is separated from the received codeword and accepted.
- If the syndrome is non-zero, it indicates that the data was corrupted during transmission and is discarded. The receiver usually requests a retransmission (ARQ).

Example:

Consider a dataword of 1001 is to be transmitted and a divisor agreed upon by both the transmitter and receiver is 1011.

Step 1: Append Zeros to the Dataword

The number of zeros to append is equal to $(n - 1)$, where n is the number of bits in the divisor.

- Divisor = 1011 (4 bits).
- Zeros to append = $4 - 1 = 3$ zeros.
- Augmented Dataword = 1001000

Step 2: Binary Division (Modulo-2)

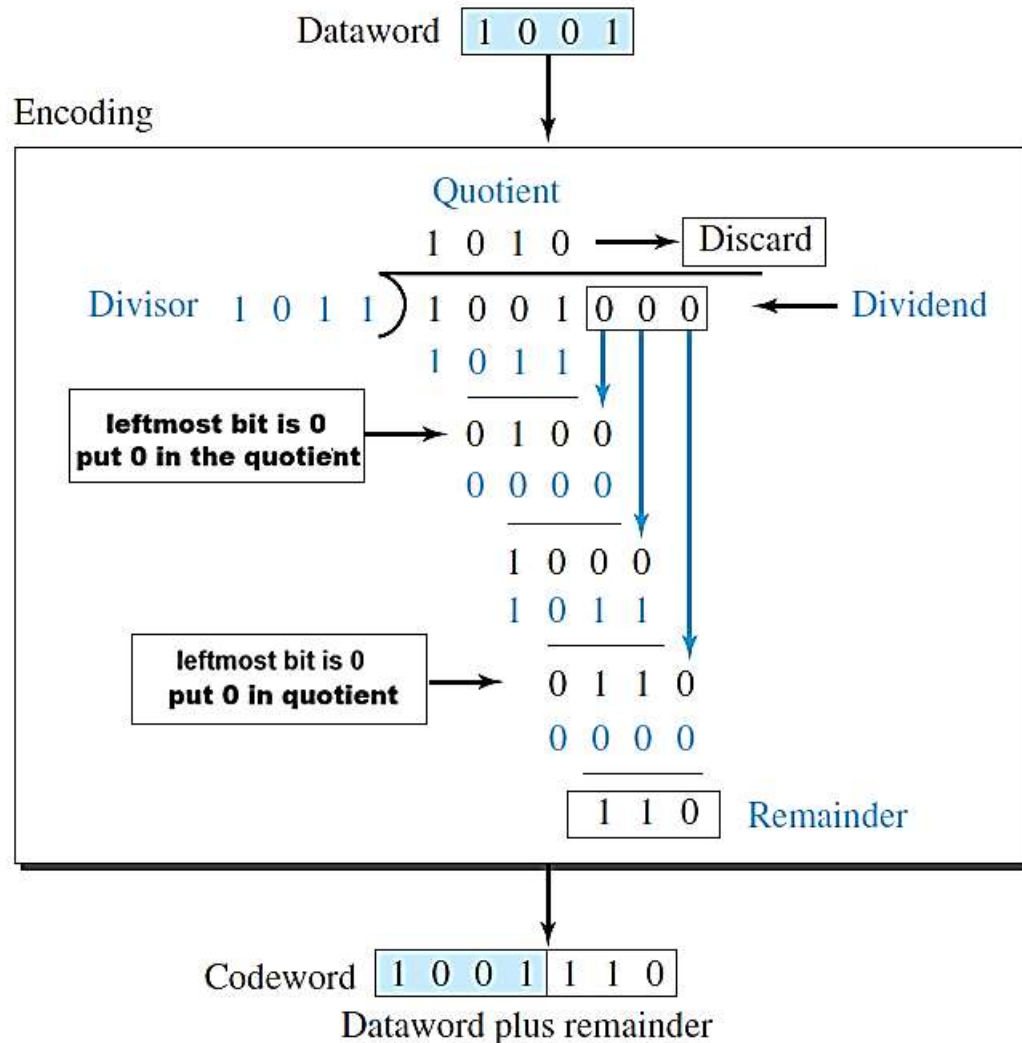
Division Process:

1. Divide 1001000 by 1011:

- Since leftmost bit of (first 4 bits) 1001 is 1, we put a 1 in the quotient and $1001 \text{ XOR } 1011 = 0010$
- Bring down the next bit (0): New value = 0100
- Since leftmost bit of the current remainder 0100 is 0, we put a 0 in the quotient and $0100 \text{ XOR } 0000 = 0100$
- Bring down the next bit (0): New value = 1000

- Since leftmost bit of the current remainder 1000 is 1, we put a 1 in the quotient and $1000 \text{ XOR } 1011 = 0011$
- Bring down the last bit (0): New value = 0110
- Since leftmost bit of the current remainder 0110 is 0, we put a 0 in the quotient and $0110 \text{ XOR } 0000 = 0110$

2. The final remainder is **110**.



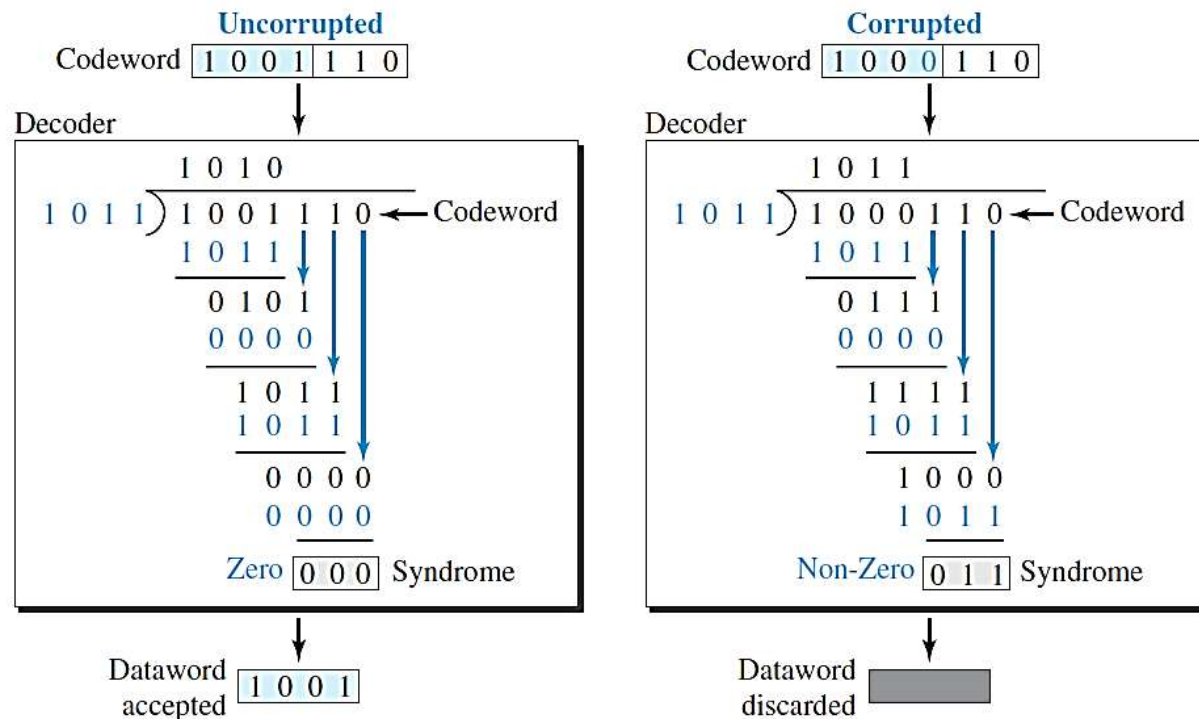
Step 3: Form the Codeword

The remainder (CRC) is appended to the original dataword.

- *Original Dataword:* 1001
- *CRC Bits:* 110

- **Transmitted Codeword:** (Append the CRC code 110 to the original dataword 1001) 1001110

Following figure shows two cases at the receiver. The receiver divides the received codeword by the same divisor 1011. The left-hand figure shows the value of the syndrome when no error has occurred; the syndrome is 000. The right-hand part of the figure shows the case in which there is a single error. The syndrome is not all 0s (it is 011).



Advantages of CRC over Parity Check

CRC offers several critical advantages over simple parity checking:

- **Burst error detection:** CRC can detect burst errors of any length up to the degree of the generator. A 32-bit CRC (CRC-32) can detect all burst errors of up to 32 bits in length.
- **All single and double-bit errors:** CRC detects all single-bit errors and all isolated double-bit errors.
- **All odd numbers of errors:** With appropriate generator selection, CRC detects any odd number of bit errors.
- **High reliability:** CRC-32 has a probability of less than 1 in 4 billion of failing to detect a random error pattern.

Applications of CRC

- Ethernet
- Wireless communication
- Storage devices

5.9.5 Comparison of Error Detecting Techniques

Parameter	Parity (VRC/LRC)	Checksum	CRC
Redundancy added	1 bit per byte (VRC)	16 or 32 bits	8 to 32 bits
Single-bit detection	Yes	Yes	Yes
Burst error detection	Limited (LRC helps)	Limited	Excellent
Mathematical basis	Bit counting	Arithmetic sum	Polynomial division
Complexity	Very low	Low	Moderate
Applications	Simple links, keyboards	IP, UDP, TCP headers	Ethernet, Wi-Fi, USB, ZIP

5.10 Error Correction Techniques - Hamming Code

In digital communication systems, errors may occur during transmission due to noise and interference. To ensure reliable data transmission, error correction techniques are used. Hamming code is a widely used error correction method that can detect and correct single-bit errors without retransmission. The other error correction techniques include block codes and convolutional codes.

Hamming code is a linear block code that adds redundant parity bits to the data bits at specific positions in such a way that the receiver can identify and correct a single-bit error automatically.

Principle of Hamming Code

- Parity bits are inserted at placed at positions that are powers of two, i.e., 1, 2, 4, 8, etc.
- The data bits are placed at positions 3, 5, 6, 7, 9, 10, 11...
- Each parity bit checks a specific combination of data bits.

- By examining parity checks at the receiver, the position of the erroneous bit can be identified.

Example: Hamming (7, 4) Code

Encoding Process at transmitter: Suppose the 4-bit data 1011 is to be transmitted.

- Step 1: Position the bits
 - Total bits = 7 (4 data + 3 parity)
 - Positions: 1 2 3 4 5 6 7
 - Type: P1 P2 D1 P4 D2 D3 D4
- Step 2: Place data bits
 - D1=1, D2=0, D3=1, D4=1
 - Positions after placing data: P1 P2 1 P4 0 1 1
- Step 3: Calculate parity bits (even parity)
 - P1 (covers positions 1,3,5,7): bits: P1, 1, 0, 1
1s count (excluding P1): $1 + 0 + 1 = 2$ (even) $\rightarrow P1 = 0$
 - P2 (covers positions 2,3,6,7): bits: P2, 1, 1, 1
1s count (excluding P2): $1 + 1 + 1 = 3$ (odd) $\rightarrow P2 = 1$
 - P4 (covers positions 4,5,6,7): P4, 0, 1, 1
1s count (excluding P4): $0 + 1 + 1 = 2$ (even) $\rightarrow P4 = 0$
- Step 4: Final encoded word
 - Insert parity bits:
 - Position: 1 2 3 4 5 6 7
 - Bits: 0 1 1 0 0 1 1
 - Encoded Hamming code: 0110011

Error detection and correction at the receiver: Now, suppose during transmission, the 5th bit flips (0 $\rightarrow$ 1) hence the received word will be 0110111 (error at position 5).

- Step 1: Recalculate parity at receiver
 - C1: (P1 checks positions 1,3,5,7): bits 0, 1, 1, 1
1s count: $0 + 1 + 1 + 1 = 3$ (odd) $\rightarrow C1 = 1$ (error)
 - C2: (P2 checks positions 2,3,6,7): bits 1, 1, 1, 1
1s count: $1 + 1 + 1 + 1 = 4$ (even) $\rightarrow C2 = 0$ (No error)
 - C3: (P4 checks positions 4,5,6,7): bits 0, 1, 1, 1
1s count: $0 + 1 + 1 + 1 = 3$ (odd) $\rightarrow C3 = 1$ (error)
- Step 2: Form syndrome word
 - Syndrome = C3 C2 C1 = 1 0 1 (binary)
- Step 3: Convert to decimal
 - $101_2 = 5_{10} \rightarrow$ Error is at position 5

- Step 4: Correct the error
 - Flip bit at position 5: $1 \rightarrow 0$
 - Corrected word: 0110011 (original encoded word)
- Step 5: Extract data
 - Remove parity bits (positions 1,2,4):
 - Original data: 1011

Advantages

- Can correct single-bit errors
- Can detect (but not correct) double-bit errors
- Systematic and relatively simple to implement

Limitations

- Cannot correct multiple errors
- Adds overhead (3 parity bits for 4 data bits = 43% overhead)

Applications

- Computer memory (ECC RAM)
- Satellite communication
- Digital data storage systems
- Computer memory systems
- Digital communication systems

5.11 Error Correction Strategies — FEC and ARQ

Beyond individual coding techniques, error control in communication systems follows one of two broad strategies: Forward Error Correction (FEC) and Automatic Repeat reQuest (ARQ). These two approaches represent fundamentally different philosophies for handling errors.

5.11.1 Forward Error Correction (FEC)

Forward Error Correction is an approach in which the sender adds sufficient redundancy to the transmitted data so that the receiver can detect and correct errors entirely on its own, without any communication back to the sender. The correction happens entirely at the receiver end, using the redundant bits that were embedded in the transmitted frame. Hamming code is one example of an FEC technique. More advanced codes used in practice include Reed-Solomon codes, Turbo codes, and Low-Density Parity-Check (LDPC) codes.

FEC is preferred in situations where:

- **Round-trip delay is large:** In satellite communication, a signal may take 270 ms or more to travel one way. Waiting for a retransmission request would waste enormous amounts of time.

- **Return channel is unavailable or unreliable:** In broadcast systems (radio, digital TV), there is no feedback channel from the receiver to the transmitter.
- **Real-time delivery is essential:** In voice calls and live video streaming, a delayed packet is useless; the receiver must recover the data immediately or discard it.

5.11.2 Automatic Repeat reQuest (ARQ)

Automatic Repeat reQuest is an error correction strategy that relies on retransmission. When the receiver detects an error (using techniques such as parity, checksum, or CRC), it sends a negative acknowledgment (NAK) back to the sender, requesting that the erroneous frame be retransmitted. If the frame arrives correctly, the receiver sends a positive acknowledgment (ACK). The sender keeps a copy of every transmitted frame until it receives an ACK, and retransmits any frame for which a NAK is received.

ARQ is preferred in situations where:

- **Round-trip delay is small:** In LAN or broadband Internet connections, the delay for a retransmission request is negligible, making ARQ very efficient.
- **Error rate is low:** If errors are rare, ARQ adds very little overhead; the channel capacity is not wasted on redundancy.
- **Data integrity is critical:** File transfers, web browsing, and database transactions cannot tolerate even a single corrupted bit; retransmission ensures perfect delivery.

5.12 Comparison: Error Detection vs Error Correction

Parameter	Error Detection	Error Correction
Purpose	Detects errors	Detects and corrects errors
Correction	Not possible	Possible
Retransmission	Required	Not required
Hardware	Simple	Complex
Bandwidth Usage	Low (few extra bits)	High (many redundant bits)
Delay	Higher (due to retransmission)	Lower (immediate correction)
Examples	Parity, Checksum, CRC	Hamming Code, Convolutional Codes

Review Question

Basic Modulation techniques (AM, FM, PM)

3 Marks Questions:

1. Define modulation and explain why it is necessary in telecommunication systems.
2. Define carrier signal, modulating signal and modulated signal.
3. What is modulation index? Explain its significance in AM.
4. Define frequency deviation in the context of FM and state its unit.
5. Define AM, FM and PM.
6. List the advantages and disadvantages of: (a) AM, (b) FM, (c) PM.
7. Differentiate between AM and FM.
8. State the applications of AM, FM and PM.

5 Marks Questions:

9. Explain the principle of Amplitude Modulation with a neat block diagram of AM transmitter and waveforms.
10. Explain the principle of Frequency Modulation with waveforms.
11. Explain the principle of Phase Modulation with waveforms.
12. Compare AM, FM and PM (any five points).

8 Marks Questions:

13. Explain Amplitude Modulation in detail with: definition, block diagram of AM transmitter, waveform, advantages, disadvantages and applications.
14. Explain Frequency Modulation in detail with: definition, waveform, advantages, disadvantages and applications.
15. Explain Phase Modulation in detail with: definition, waveform, advantages, disadvantages and applications.

Multiplexing: FDM, TDM

3 Marks Questions

16. Define multiplexing. State the need for multiplexing in communication systems.

17. What is a multiplexer (MUX) and demultiplexer (DEMUX)? State their functions.
18. What is meant by channel capacity in multiplexing?
19. Define FDM and TDM.
20. What are guard bands in FDM? Why are they required?
21. What is crosstalk in FDM? How can it be minimized?
22. What are time slots in TDM? Explain their significance.
23. Why is synchronization critical in TDM systems?
24. State advantages and disadvantages of FDM.
25. State advantages and disadvantages of TDM.
26. List the applications of FDM and TDM.
27. What is meant by synchronous and asynchronous TDM?
28. Explain how multiplexing helps in the efficient utilization of available bandwidth.
29. What is synchronization in TDM? Why is it important?

5 Marks Questions

30. Explain the principle of Frequency Division Multiplexing (FDM) with neat block diagram.
31. Explain the principle of Time Division Multiplexing (TDM) with neat block diagram.
32. Compare FDM and TDM (any five points).
33. What are guard bands in FDM? Explain the role of guard bands in FDM.

8 Marks Questions

34. Explain Frequency Division Multiplexing (FDM) in detail with: Definition, Block diagram, Working principle, Advantages, Disadvantages, Applications
35. Explain Time Division Multiplexing (TDM) in detail with: Definition, Block diagram, Frame structure, Working principle, Advantages, Disadvantages, Applications
36. With neat block diagrams, explain the complete FDM system including both transmitter and receiver. Discuss how guard bands prevent crosstalk.

37. With neat block diagrams, explain the complete TDM system including both multiplexer and demultiplexer. Discuss the importance of synchronization.

Error detection and correction techniques

3 Marks Questions

38. Define error detection and error correction. State the difference between them.
39. What is the need for error control in digital communication systems?
40. Explain single-bit error and burst error with example.
41. What is redundancy in error control coding? Why is it necessary?
42. What is VRC? State advantages and disadvantages of VRC.
43. What is CRC? State its advantage over parity check.
44. State advantages, disadvantages and applications of hamming code.
45. What is Forward Error Correction (FEC)?
46. What is Automatic Repeat reQuest (ARQ)?

5 Marks Questions

47. What is a burst error? How does LRC help in detecting burst errors compared to VRC?
48. Explain the basic working principle of CRC. State its advantage over parity check. State its applications.
49. Compare error detecting methods parity check, checksum method and CRC method on the parameters: redundancy added, burst error detection, mathematical basis, complexity and applications.
50. Differentiate between error detection and error correction techniques.

8 Marks Questions

51. Explain the parity checking method for error detection. How does two-dimensional parity (LRC) improve upon single parity (VRC)?
52. Describe the checksum method of error detection with suitable example. State its advantages, limitations and applications.
53. Describe the CRC method of error detection with suitable example.
54. Explain the various error detection techniques — Parity Check, Checksum, and CRC.

55. Explain the Hamming code in detail using suitable example.
56. Explain Forward Error Correction (FEC) and Automatic Repeat reQuest (ARQ). State where each is preferred.